

Double Counting

JISHNU GHOSH

June 2025

§1 Problems

§1.1 Level 1

1. There are n people and m committees. Each person belongs to exactly 10 committees, and each committee has exactly 5 members. What is the relation between m and n .
2. A set of 8 problems was prepared for an exam. Each student was given 3 of them. No two students received more than one common problem. What is the maximum number of students?

§1.2 Level 2

1. Two hundred students participate in a mathematical contest. They had six problems to solve. It is known that each problem was correctly solved by at least 120 participants. Prove that there must be two participants such that every problem was solved by at least one of these two students.
2. In a competition, there are m contestants and n judges, where $n \geq 3$ and n is odd. Each judge rates each contestant as either pass or fail. Suppose k is a number such that the ratings of each pair of judges coincides for at most k contestants. Show that

$$k \geq \frac{m(n-1)}{2n}.$$

3. There are n points in the plane such that no three of them are collinear. Prove that the number of triangles whose vertices are chosen from these n points and whose area is 1 is not greater than $\frac{2}{3}(n^2 - n)$.
4. Prove that for all positive integer n , the following equality holds:

$$\sum_{i=0}^{\lfloor n/2 \rfloor} 2^{n-2i} \binom{n}{n-2i} \binom{2i}{i} = \binom{2n}{n}.$$

5. Prove that

$$\sum_{k=0}^n 2^k \binom{n}{k} \binom{n-k}{\lfloor \frac{n-k}{2} \rfloor} = \binom{2n+1}{n}.$$

§1.3 Level 3

1. Let n and k be positive integers and let S be the set of n points in the plane such that
 - i. No three points are collinear.
 - ii. For every point P of S there are at least k points of S equidistant from P .

Prove that:

$$k < \frac{1}{2} + \sqrt{2n}.$$

2. Let n be an odd integer greater than 1 and let $c_1, c_2, \dots, c_n$ be integers. For each permutation $a = \{a_1, a_2, \dots, a_n\}$ of $\{1, 2, \dots, n\}$, define $S(a) = \sum_{i=1}^n a_i c_i$. Prove that there exist permutations $a \neq b$ of $\{1, 2, \dots, n\}$ such that $n!$ is a divisor of $S(a) - S(b)$.
3. Let X be a set with n elements. Given $k > 2$ subsets of X , each with at least r elements, show that we can always find two of them whose intersection has at least

$$r - \frac{nk}{4(k-1)}$$

elements.

4. There are n people at the party. Prove that there are two people such that of the remaining $n-2$ people, there are at least $\lfloor n/2 \rfloor - 1$ of them, each of whom either knows both or neither of the two.
5. Let $A_n = \{1, 2, \dots, n\}$. Let S_n be the set of permutations of $\{1, 2, \dots, n\}$. We consider each permutation as a bijective function from A_n to A_n . For $\sigma \in S_n$, $m \in A_n$ is called a fixed point of σ if $\sigma(m) = m$. Let $d(n, k)$ denote the number of elements of S_n having exactly k fixed points. Then prove that
 - a. $\sum_{k=0}^n d(n, k) = n!$
 - b. $\sum_{k=0}^n k d(n, k) = n!$
 - c. $\sum_{k=0}^n k(k-1) d(n, k) = n!$
 - d. $\sum_{k=0}^n r! \binom{k}{r} d(n, k) = n!$ for all $0 \leq r \leq n$.
6. A school has n students. Each student can participate in any number of classes that he wants. Every class has at least two students participating in it, and if two different classes have at least two common students, then the number of students in these two classes is different. Prove that the number of classes is not greater than $(n-1)^2$.

7. A binary word of length n is a sequence of 0's and 1's of length n . The set $\{0, 1\}^n$ is the set of all binary words of length n . Let S be a subset of $\{0, 1\}^n$ with the following property: for every pair of distinct elements $x = x_1x_2 \cdots x_n$ and $y = y_1y_2 \cdots y_n$, we have that x and y differ in at least 3 positions. Show that S has at most $\frac{2^n}{n+1}$ elements.

§1.4 Level 4 (Counting with Graphs)

1. Let G be a graph with n vertices $v_1, v_2, \dots, v_n$. Let d_i be the degree of v_i , E be the set of edges and $|E| = k$. Prove the following results:
 - i. $\sum_{i=1}^n d_i = 2k$
 - ii. $\sum_{i=1}^n d_i^2 \geq \frac{4k^2}{n}$
 - iii. $\sum_{i=1}^n \binom{d_i}{2} \geq \frac{2k^2}{n} - k$
 - iv. $\sum_{v_i v_j \in E} (d_i + d_j) = \sum_{i=1}^n d_i^2$
2. [**Mantel's Theorem**] Prove that the number of edges of a triangle free graph on n vertices is at most $\lfloor n^2/4 \rfloor$.
3. Consider a set of $2n$ points in space, $n > 1$. Suppose they are joined by at least $n^2 + 1$ segments. Show that at least one triangle is formed. Show that for each n , it is possible to have $2n$ points joined by n^2 segments without any triangles being formed.
4. Show that a graph with n vertices and k edges has at least $k(4k - n^2)/3n$ triangles.
5. Let G be a graph with E edges, n vertices and no 4-cycles. Show that

$$E \leq \frac{n}{4}(1 + \sqrt{4n - 3}).$$

6. Suppose 91 distinct positive integers greater than 1 are given such that there are at least 456 pairs, among them which are relatively prime. Show that one can find four integers a, b, c, d among them such that $\gcd(a, b) = \gcd(b, c) = \gcd(c, d) = \gcd(d, a) = 1$.
7. There are n distinct points in the plane. Prove that the number of point pairs with unit distance is smaller than $\frac{n}{4} + \frac{\sqrt{2}}{2}n^{3/2}$.